Priority-Aware and Load-Balanced Multi-UAV Data Collection for IoT Networks Using Clustering and Route Optimization

1. Research Motivation

The rapid growth of Internet of Things (IoT) deployments has resulted in large-scale sensor networks spread across wide geographical regions. Collecting data from these sensors using fixed infrastructure can be expensive and inefficient, particularly in remote or inaccessible environments. Unmanned Aerial Vehicles (UAVs) provide a flexible and cost-effective solution by acting as mobile data collectors.

However, when multiple UAVs are deployed simultaneously, several challenges arise:

- Uneven distribution of sensing tasks among UAVs.
- Increased mission completion time due to inefficient routing.
- Redundant coverage of sensing regions.
- Delayed collection of critical sensor data.
- Increased energy consumption and travel distance.

Most existing approaches focus primarily on clustering and route optimization while assuming that all sensors have equal importance. In practical IoT systems, however, different sensors generate data with varying levels of urgency and significance.

This motivates the need for a data collection framework that not only optimizes UAV routes but also prioritizes important sensor data while maintaining balanced workloads among UAVs.

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2. Proposed Research Objective

The objective of this research is to develop a multi-UAV data collection framework that:

- Maximizes sensor coverage.
- Minimizes total travel distance.
- Reduces mission completion time.
- Balances workload among UAVs.

- Prioritizes the collection of critical sensor data.
- Minimizes redundant coverage and inefficient resource utilization.

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3. System Model

Environment

A two-dimensional sensing field is considered.

- Multiple IoT sensors are randomly deployed.
- Multiple UAVs are available for data collection.
- Sensors are static.
- UAVs communicate with sensors within a predefined communication range.
- Dynamic obstacles, collision avoidance, and complex flight dynamics are not considered.

This simplified environment allows the research to focus on cooperative data collection and optimization rather than navigation complexity.

Sensor Model

Each sensor is characterized by:

- Location coordinates (x, y)
- Data generation information
- Priority level

Priority categories:

- High Priority
- Medium Priority
- Low Priority

Examples:

- Fire detection sensors → High Priority
- Gas leakage sensors → High Priority
- Environmental monitoring sensors → Medium Priority
- Logging and archival sensors → Low Priority

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4. Proposed Framework

Step 1: Sensor Deployment

IoT sensors are deployed across the sensing region.

Each sensor is assigned a priority value based on the importance of its generated data.

Step 2: Priority-Aware Clustering

Instead of clustering sensors solely based on geographic distance, clustering incorporates sensor priority information.

The clustering process groups sensors into regions while ensuring that critical sensors are appropriately represented during cluster formation.

Output:

- Cluster 1
- Cluster 2
- Cluster 3
- ...
- Cluster K

Step 3: Initial UAV Assignment

Each UAV is assigned to a cluster for data collection.

This provides an initial distribution of sensing tasks among UAVs.

Step 4: Load-Balanced Reassignment

After initial assignment, cluster workloads are evaluated.

If one UAV receives significantly more sensors than others, boundary sensors are reassigned to neighboring clusters.

The goal is to achieve:

- Balanced sensor distribution

- Reduced mission completion time
- Improved resource utilization

Step 5: Route Optimization

Within each cluster, an optimized route is generated for the assigned UAV.

Possible routing methods include:

- Nearest Neighbor Search
- Traveling Salesman Problem (TSP)-based optimization
- Heuristic route generation

The route minimizes travel distance while maintaining efficient coverage.

Step 6: Priority-Aware Data Collection

During data collection, UAVs prioritize sensors according to their importance.

The routing strategy favors:

High Priority Sensors → Medium Priority Sensors → Low Priority Sensors

This ensures that critical information is collected earlier during the mission.

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5. Novelty of the Proposed Work

The novelty does not come from introducing an entirely new UAV architecture, but from improving existing cluster-based multi-UAV data collection frameworks through priority-aware and workload-aware optimization.

Existing Approaches

Most existing studies focus on:

- Sensor clustering
- UAV assignment
- Route optimization
- Energy minimization

However, many assume:

- All sensors have equal importance.
- Fixed cluster assignments.
- No consideration of balanced workloads.
- No mechanism for prioritizing critical sensor data.

Proposed Contribution

This work introduces:

Priority-Aware Clustering

Sensor importance is incorporated into the clustering process rather than relying solely on geographic proximity.

Load-Balanced UAV Assignment

Cluster workloads are dynamically balanced through reassignment mechanisms.

Priority-Aware Route Planning

Important sensors are collected earlier during the mission.

Integrated Optimization Framework

The proposed framework jointly considers:

- Coverage efficiency
- Workload balancing
- Sensor priority
- Route optimization

within a unified multi-UAV data collection architecture.

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6. Research Workflow

IoT Sensor Deployment

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Priority Assignment

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Priority-Aware Clustering

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Initial UAV Assignment

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Load-Balanced Reassignment

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Route Optimization

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Priority-Aware Data Collection

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Performance Evaluation

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7. Performance Evaluation Metrics

The proposed framework will be evaluated using:

Coverage Ratio

Percentage of sensors successfully collected.

Total Travel Distance

Total distance traveled by all UAVs.

Energy Consumption

Estimated using a distance-based energy model.

Mission Completion Time

Time required to collect data from all sensors.

Workload Fairness

Measure of how evenly sensing tasks are distributed among UAVs.

Priority Collection Efficiency

Percentage of high-priority sensors collected early in the mission.

Priority Satisfaction Score

Weighted metric reflecting successful collection of critical sensor data.

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8. Expected Outcome

The proposed framework is expected to:

- Improve collection efficiency for critical sensor data.
- Reduce workload imbalance among UAVs.
- Lower overall mission completion time.
- Reduce travel distance and energy consumption.
- Provide a practical and scalable solution for multi-UAV IoT data collection in large sensing environments.

The final result will be a lightweight, simulation-based, multi-UAV data collection framework suitable for large-scale IoT deployments and capable of demonstrating measurable improvements over conventional clustering and routing approaches.